

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8506**

AY:	2025-26	Session:	08-10-2025	Date:
Sem:	1	Slot:	10:00am - 11:00am	C. Code:
Exam Type:	Written	Block:	401	CS413
		Course Section:	1	2022600054

L_29_511_986_1_1505_38352



Name of the candidate. Shah Jai Bhavesh

Candidate's UID number:

2	0	2	2	6	0	0	0	5	4
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Examination class room number:

4	0	1
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Invigilator's signature with date:

<u>[Signature]</u> 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : 7

PROGRAM: CSE AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday and 8/10/25

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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1a

gradient update Rule

$$w_{t+1} = w_t - \eta \nabla w_t$$

$$b_{t+1} = b_t - \eta \nabla b_t$$

where $\nabla w_t = \frac{\partial L(w, b)}{\partial w}$ and $\nabla b_t = \frac{\partial L(w, b)}{\partial b}$

$$f(w) = \frac{1}{2} \|xw - y\|^2$$

$$\nabla w_t = \frac{\partial L(w, b)}{\partial w} = \frac{\partial}{\partial w} \frac{1}{2} (f(w) - y)^2$$

$$= \frac{1}{2} * (2 * (f(w) - y) * \frac{\partial}{\partial w} (f(w) - y))$$

$$= (f(w) - y) * \frac{\partial}{\partial w} (f(w))$$

$$= (f(w) - y) * \frac{\partial}{\partial w} \frac{1}{2} (xw - y)^2$$

$$= (f(w) - y) * \frac{1}{2} * 2 * (xw - y) * \frac{\partial}{\partial w} (xw - y)$$

$$= (f(w) - y) * (xw - y) * \frac{\partial}{\partial w} xw$$

$$\nabla w_t = (f(w) - y) * (xw - y) * x$$

~~$$\nabla w_t = (f(w) - y) * (2f(w) * \sqrt{2f(w)}) * x$$~~

if there are n point

~~$$\sum_{i=1}^n (f(xw_i) - y_i) (\sqrt{2f(w_i)}) * x_i$$~~

~~$$\nabla w_t = (f(xw) - y) * (xw - y)$$~~

$$\nabla w_t = (f(w) - y) (\sqrt{2f(w)}) * x$$

[illegible]

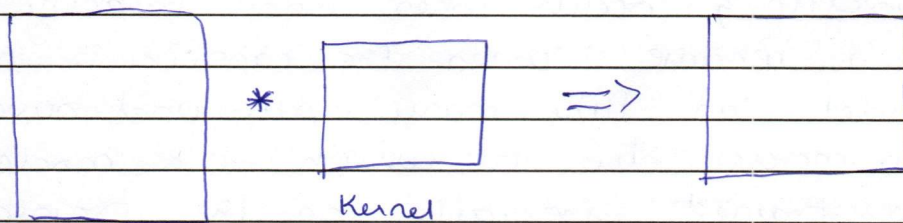
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Q

2a



input image
 $H \times W \times C$

Kernel
 $F \times F$
No. of filters = K

$$W_2 \text{ (Output image width)} = \frac{W - F + 2P}{S} + 1$$

$$H_2 \text{ (Output image height)} = \frac{H - F + 2P}{S} + 1$$

$$D_2 \text{ (Output image depth)} = K$$

$$\therefore \text{Output image volume} = W_2 \times H_2 \times D_2$$

$$\text{parameter} = F \times F \times K$$

$$\text{Computational complexity} = 3$$

Depth w_1

Depthwise separable convolutions are critical in modern architecture because the accuracy which it can attain is very good.

~~There will be lesser parameters to be attained~~
The model can go in depth select all the input important feature and then it will display the output which will be ^{very} accurate.

Parameter efficiency in

$$\text{In 1st convolutional layer} = F \times F \times K$$

depthwise separable convolution would be ^{more} ~~less~~ ^{because} ~~more~~ ^{more} ~~less~~ number of parameters would be ~~more~~ ^{less} parameter would be explored so it's efficiency is good than standard 2D convolution.

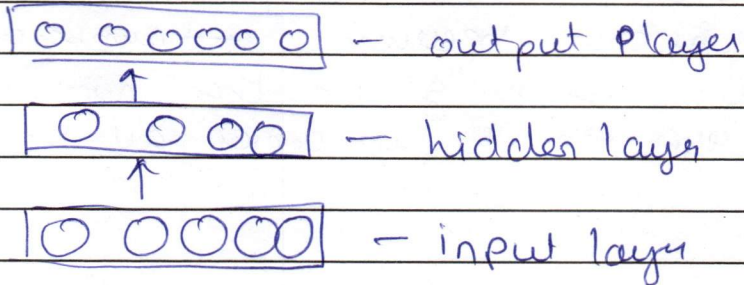


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2b) Undercomplete autoencoders means the hidden layer is smaller than the input layer.

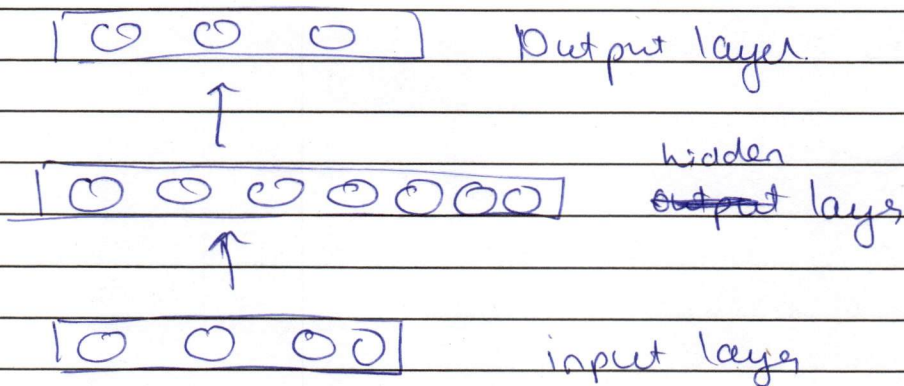


$$MSE = \frac{1}{n} \sum_{i=1}^N (\hat{y}_i - y_i)^2$$

PCA

Overcomplete Autoencoders

The hidden layer is greater than the input layer.



So in overcomplete encoder the hidden layer copies all the input from input layer it does not require to compress and select only important feature as the number of units in hidden layer are ~~less~~ in greater number than the input layer.

So overcomplete layer copies all the input and can process everything and then forward the data.

So in regularization the features are either dropped or reduces their noise basically it helps to compress and helps to find

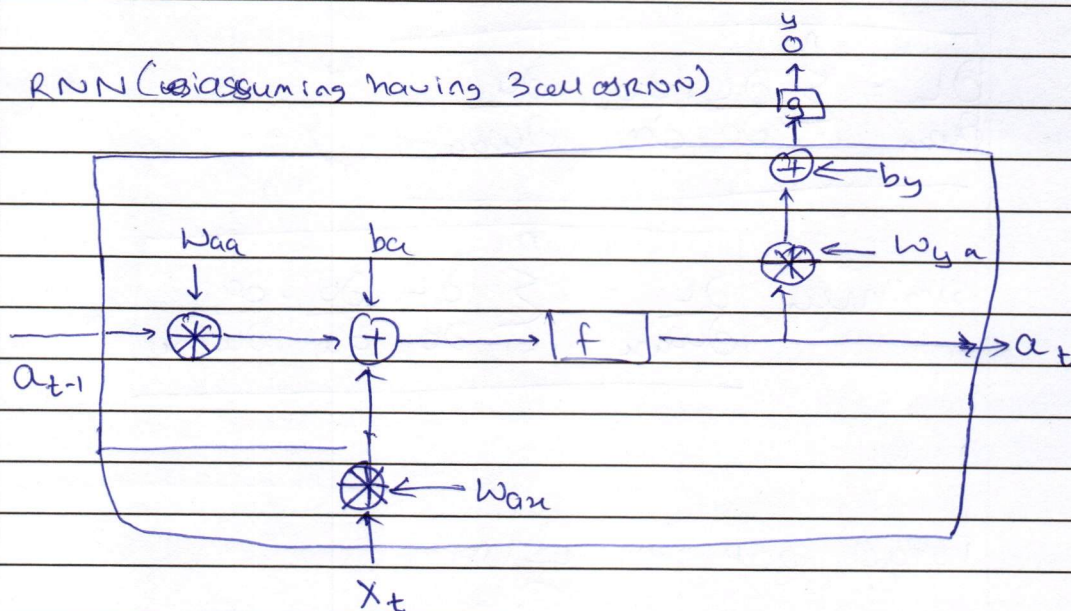
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3a

RNN (assuming having 3 cell as RNN)


 $\otimes \rightarrow$ multiplication

 $\hat{y}$ = predicted output
 y = actual output

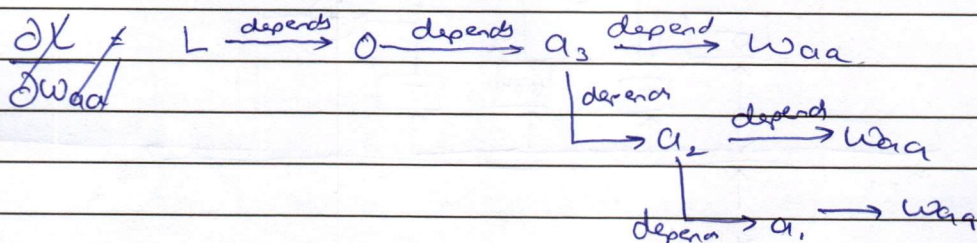
loss function $\rightarrow$ categorical cross entropy = $-\sum_{i=1}^n y_i \log(\hat{y}_i)$

 $f = \tanh$ $g = \text{softmax}$
 $w_{aa} = \text{weight matrix}$
 $w_{ax} = w_{ya} = \text{weight matrix}$

Back propagation

$$\frac{\partial L}{\partial w_{ya}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_{ya}} = (y - \hat{y}) \cdot a_t$$

$$w_{t+1} = w_t - \alpha [(y - \hat{y}) \cdot a_t]$$



$$\frac{\partial L}{\partial w_{aa}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_3} \cdot \frac{\partial a_3}{\partial w_{aa}} + \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_2} \cdot \frac{\partial a_2}{\partial w_{aa}} + \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_1} \cdot \frac{\partial a_1}{\partial w_{aa}}$$

So

$$\frac{\partial L}{\partial w_{aa}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_3} \cdot \frac{\partial a_3}{\partial w_{aa}} + \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_2} \cdot \frac{\partial a_2}{\partial w_{aa}} + \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_1} \cdot \frac{\partial a_1}{\partial w_{aa}}$$

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$$\frac{\partial L}{\partial w_{aa}} = \sum_{i=1}^n \frac{\partial L}{\partial o} \cdot \frac{\partial o}{\partial a_i} \cdot \frac{\partial a_i}{\partial w_{aa}}$$

similarly

$$\frac{\partial L}{\partial w_{ax}} = \sum_{i=1}^n \frac{\partial L}{\partial o} \cdot \frac{\partial o}{\partial a_i} \cdot \frac{\partial a_i}{\partial w_{ax}}$$

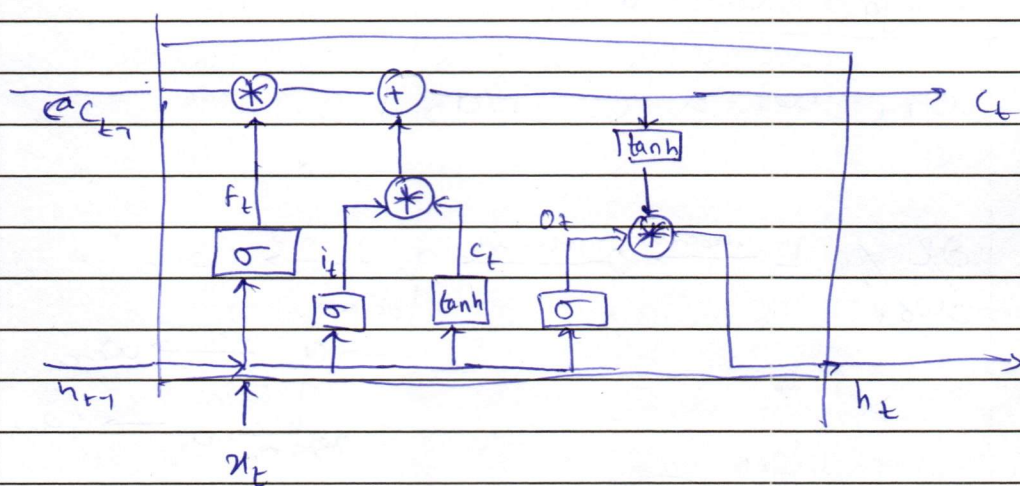
Forget gate in LSTM

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

$$= \sigma(w_{fh} * h_{t-1} + w_{fx} * x_t + b_f)$$

When $b_f h \Rightarrow$ Short memory state

So forget gate will allow only those information which are invalid or which is not going to be in use.



So basically forget gate has the responsibility to decide which information to retain or to delete.

So here there are less iterations ^{or travelling backwards} so the value stored in unit won't be negligible ^{or lost} so here it is more efficient to remember so in this case vanishing gradient is more suitable and it is better than RNN.


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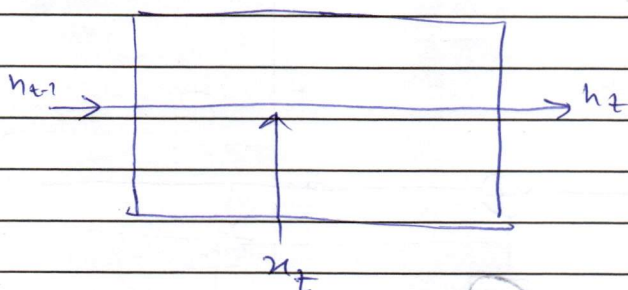
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 3b In GRU there is only 1 state (h)

In GRU there are 2 gate

- 1) Update Gate
- 2) reset gate



candidate value

$$h_t = U_t * \tilde{C}_t + (1 - U_t) * h_{t-1}$$

$$\tilde{C}_t = \tanh(W_c [x_t, h_{t-1}] + b_c)$$

$$U_t = \sigma(W_u [h_{t-1}, x_t] + b_u)$$

$$r_t = \sigma(W_r [h_{t-1}, x_t] + b_r)$$

So the update gate only takes input as 0 and 1

 so when $U_t = 0$

$$h_t = (1 - 0) * h_{t-1} =$$

$$\therefore h_t = h_{t-1}$$

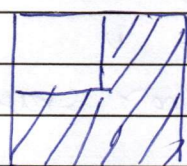
Here the information will be retained.

 When $U_t = 1$

$$h_t = \tilde{C}_t + 0 \therefore h_t = \tilde{C}_t$$

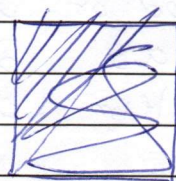
 role of candidate value is to allow new information to store so when $U=1$ new information will be stored

eg



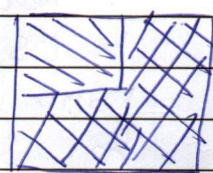
old information

+



new information

=>



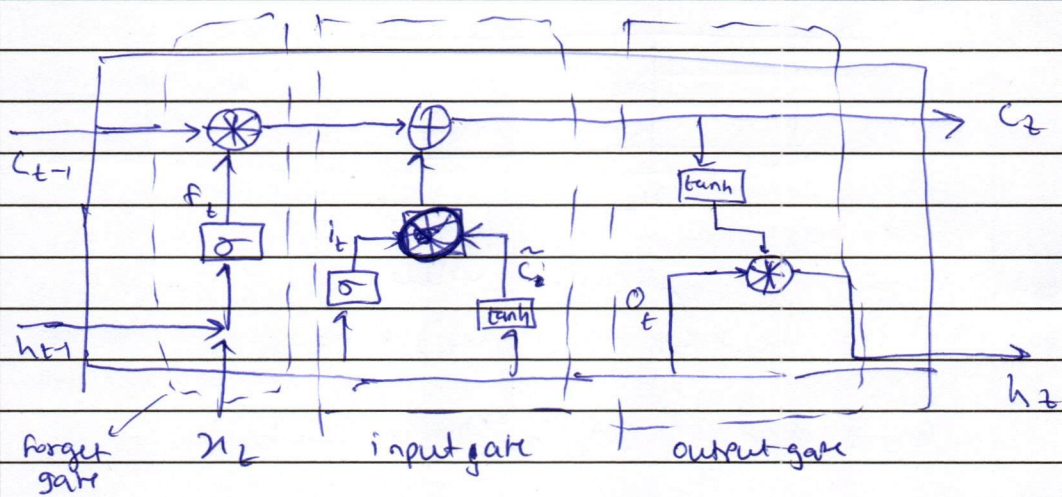
new information

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So basically new information block
will it retain how as the information
and rest it will interpret and create
an information.

LSTMs



↗ candidate value

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

$$O_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

$$h_t = O_t * \tanh(C_t)$$

So in LSTM there are 3 gates forget gate
input gate, output gate

So LSTM has to go through all of these
gate and then compute which information
it needs to retain or update.

In LSTM there are 2 state
1 state is for long term and other state is
for short term

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In GRU there is only 1 state which includes both long term and short term.

Due to which GRU has less parameters than LSTM

1) GRU has less state

2) GRU has less number of gate

GRU is computationally fast than LSTM because as number so GRU can be used in Stock market analysis as it will provide faster response.

GRU can be used in completing the sentence or filling a missing blank type scenarios.

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14

[illegible]

[illegible]

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